Virtual Stub Column Compression Test

Formulas used, where each one comes from, and where and how it produces the output · app page **Stub Column Test** (files `stub.py`, `stub_page.py`) · repository vectorverse11/model_spm

Source key

Key	Meaning
A	Your procedure / Excel sheet (EN 15512 Annex A.2.1)
B	Eurocode clause added by Claude (EN 1993-1-1 / 1-3 / 1-5)
C	Thin-walled section theory (textbook, no code clause)
D	Modelling choice by Claude (no standard prescribes it)

Clause numbers for EN 1993 were written from knowledge, not copied from the documents you supplied — please check them against your copies. Fixed constants: E = 210 000 N/mm², G = E / (2(1 + ν)) = 80 769 N/mm², ν = 0.3. Units: N, mm (loads shown in kN).

1 · User inputs on the Streamlit page

Group	Inputs	Used in
Specimen	W (web width), D (flange depth), t, lip length (0 = none), c (cold-formed length)	All section formulas, checks, buckling, shortening
Perforations	pitch p; holes across the web in one row n_w, width d_w; holes across each flange n_f, width d_f	5-hole check, A_{net}, effective strips
Cap / base plates	length, width, thickness of each	Projection check, cap thickness check, b
Material	f_y (typed), γ_{M0}: IS 2062 grade dropdown is reference only	ε, λ_p, ρ, A_{eff}, N_u, $\bar{\lambda}$, $A_{eff} = R_k/f_y$, $N_{b,Rd}$
Test	Alternative 1 or 2; dial least count LC (mm/div); load increment ΔN (kN)	Buckling lengths, load position, dial readings, load steps

2 · Formulas, sources, and how they produce the output

#	Formula	Source	Where in code	How it is used / what it produces
1	$c \geq 3 \cdot b_{max}$ (b_{max} = greatest flat width)	A — sheet item 1 / A.2.1.2 a)	specimen_checks()	Specimen-checks table ✓/✗ (warning only)
2	$c / p \geq 5$	A — sheet item 2 / A.2.1.2 a)	specimen_checks()	Specimen-checks table
3	$50 \leq p \leq 70$ mm	A — sheet item 2	specimen_checks()	Specimen-checks table
4	$(L_{cap} - W)/2 \geq 10$ and $(L_{base} - W)/2 \geq 10$ (along the web only)	A — sheet item 4 (per-side reading and web-only by user decision)	specimen_checks()	Specimen-checks table
5	$t_{cap} \geq 30$ mm	A — sheet item 5	specimen_checks()	Specimen-checks table
6	$b = c + 2\cdot(t_{cap} + t_{base})$	A — sheet value b = 380 (formula is Claude's reading of it)	buckling_length_b()	Buckling length shown under the plate inputs; L_F for Alternative 1
7	$b_w = W - t$; $b_f = D - t$ ($D - t/2$ without lip); $b_l = d_{lip} - t/2$	C — centreline model	widths()	Flat widths for every later formula
8	$A_g = t \cdot \Sigma b_i$; $x_g = \Sigma(A_i x_i)/A_g$	C	gross_properties()	Step-by-step §2; load position (Alt. 2); Q
9	I_y, $I_z = \Sigma \int r^2\, dA$; $I_t = \Sigma\, b\, t^3/3$	C	gross_properties()	N_{cr} (formula 18–20)
10	$\omega(s)$ sectorial coordinate → shear centre x_0, $I_w = \int\omega^2\, dA$, $i_0^2 = (I_y+I_z)/A + x_0^2$	C — Vlasov thin-walled theory	gross_properties()	Torsional / torsional-flexural N_{cr}
11	$A_{net} = A_g - (n_w d_w + 2n_f d_f)\cdot t$; x_{net}	A — 'minimum cross-section' named in procedure; formula C	net_area()	Step §2; Alt. 1 load range; proof load; start value of A_{eff} iteration
12	$\varepsilon = \sqrt{(235/f_y)}$; $\lambda_p = (b/t) / (28.4\cdot\varepsilon\cdot\sqrt{k_\sigma}) \cdot \sqrt{(\sigma/f_y)}$	B — EN 1993-1-5 §4.4(2); σ-scaling: §4.4(4) / Annex E	rho_of()	Plate slenderness at each load step and at f_y
13	$k_\sigma = 4$ (internal: web, flange with lip); $k_\sigma = 0.43$ (outstand: lip, flange without lip)	B — EN 1993-1-5 Tables 4.1 / 4.2 (ψ = 1)	elements()	Input to λ_p
14	internal: $\rho = 1$ if $\lambda_p \leq 0.673$, else $\rho = (\lambda_p - 0.22)/\lambda_p^2$	B — EN 1993-1-5 §4.4(2) eq. 4.2 (ψ = 1)	rho_of()	Effective width of web and flanges
15	outstand: $\rho = 1$ if $\lambda_p \leq 0.748$, else $\rho = (\lambda_p - 0.188)/\lambda_p^2$	B — EN 1993-1-5 §4.4(2) eq. 4.3	rho_of()	Effective width of lips
16	internal: strips 0.5ρb at each edge; outstand: ρb from supported edge; central hole removes only its overlap with the strips → b_{eff}	B (strip layout, EN 1993-1-5 Table 4.1) + D (hole-overlap rule)	_strips(), effective_section()	Effective-width table; A_{eff}
17	$A_{eff} = t\cdot\Sigma b_{eff}$; x_{eff}; I_{eff}; $W_{eff} = I_{eff}/c_{max}$; $Q = A_{eff}/A_g$	B — EN 1993-1-3 §5.5.2 / §6.1.3; Q: EN 15512	effective_section()	A_{eff}, Q, W_{eff} shown in §3; used in N_u
18	Alt. 1: $L_F = b$, $L_T = c/2$; Alt. 2: $L_F = L_T = c$ (full length)	A — your sheet (EN 15512 A.2.1.4); Alt. 2 c/2 rule removed by user decision	critical_loads()	Buckling lengths in §4
19	$N_{cr,y} = \pi^2 E I_y/L_F^2$; $N_{cr,z} = \pi^2 E I_z/L_F^2$; $N_{cr,T} = (G\cdot I_t + \pi^2 E\cdot I_w/L_T^2)/i_0^2$	B — EN 1993-1-3 §6.2.3	critical_loads()	Critical loads in §4
20	$i_0^2(N-N_y)(N-N_z)(N-N_T) - N^2 x_0^2(N-N_z) = 0$ → smallest root $N_{cr,TF}$; $N_{cr} = \min(\text{all})$	B — EN 1993-1-3 §6.2.3(7) (general cubic form)	critical_loads()	N_{cr} for $\bar{\lambda}$
21	$\bar{\lambda} = \sqrt{(A_{eff}\cdot f_y/N_{cr})}$	B — EN 1993-1-1 §6.3.1.2	failure_load()	Slenderness shown in §4
22	$\chi = 1$ if $\bar{\lambda} \leq 0.2$; else $\Phi = 0.5[1 + \alpha(\bar{\lambda}-0.2) + \bar{\lambda}^2]$, $\chi = 1/(\Phi + \sqrt{(\Phi^2-\bar{\lambda}^2)})$, $\alpha = 0.34$	B — EN 1993-1-1 §6.3.1.2(1),(4); curve b: EN 1993-1-3 Table 6.3; 0.2 limit also in your sheet	chi_of()	Reduction for buckling of the stub
23	$N_c = \chi \cdot A_{eff} \cdot f_y$	B — EN 1993-1-3 §6.1.3 / §6.2.1	failure_load()	Concentric failure load
24	Alt. 2: $e = 0$ → $N_u = N_c$	A (clamped platen, gross centroid central) + D ($e = 0$)	failure_load()	**Failure load, Alternative 2**

#	Formula	Source	Where in code	How it is used / what it produces
25	Alt. 1: $x_{load} = x_g + f(x_{net} - x_g)$, f = 0…1; $e = x_{load} - x_{eff}$; $N_u = \max_f 1/(1/N_c + \lvert e\rvert/(W_{eff} f_y))$	A (position range, 'adjusted to give max failure load') + B (EN 1993-1-3 §6.1.9 interaction) + D (sweep)	failure_load()	**Failure load, Alternative 1**
26	$\sigma = \min(N/A_{eff}(\sigma), f_y)$ iterated to convergence	D (with formulas 12–17)	_A_eff_at_load()	A_{eff} at every load step
27	Loads: $N_i = i \cdot \Delta N$, i = 0,1,… until $N \geq N_u$; last row at N_u	A (load in increments to failure) + D	run()	Rows of the readings table
28	$\delta = N \cdot c / (E \cdot A_{eff}(\sigma))$; Dial = δ / LC	D — elastic shortening	run()	Shortening (mm) and Dial (div) columns; both graphs
29	$\sigma_{lim,i} = f_y(\lambda_{lim}/\lambda_{p,i}(f_y))^2$, λ_{lim} = 0.673 / 0.748; $N_{proof} = \min \sigma_{lim} \cdot A_{net}$	D (definition) using B limits	proof_stress(), run()	Proof load metric and line on slope graph
30	Slope1 = $N_{proof}/Dial_{proof}$; Slope2 = $(N_u - N_{proof})/(Dial_u - Dial_{proof})$; Slope3 = $N_u/Dial_u$	D — layout modelled on the vLab reference	run()	Slope lines and values (kN/div)
31	$R_k = N_u$	A — 'failure load' / 'maximum load recorded'	run()	**Failure load R_k metric**
32	$A_{eff} = R_k/f_y$; if $\bar{\lambda} > 0.2$ solve $\chi(A_{eff}) \cdot A_{eff} \cdot f_y = R_k$	A — your sheet ('area adjusted if slenderness > 0.2'); equation form B / EN 15512 A.2.1.4	a_eff_from_test()	**$A_{eff} = R_k/f_y$ metric**
33	$N_{b,Rd} = R_k / \gamma_{M0}$	A — your sheet	run()	**$N_{b,Rd}$ metric**

3 · How the formulas chain into the output

Stage	Formulas	Output on the page
Inputs → checks	1–6	Specimen-checks table (✓/✗), buckling length b
Section	7–11	Step-by-step §2: widths, A_g, A_{net}, I, I_t, I_w, shear centre
Local buckling	12–17 (at σ = f_y)	Effective-width table, A_{eff}, x_{eff}, W_{eff}, Q (§3)
Stub buckling	18–23	§4: L_F, L_T, N_{cr}, $\bar{\lambda}$, χ, N_c
Failure	24 or 25	§5: failure load N_u (green banner)
Virtual loading	26–28 (12–17 at every step)	Readings table, Load vs Dial graph, CSV
Curve features	29–30	Proof load, Slope1–3 graph
EN 15512 results	31–33	Metrics R_k, A_{eff}, $N_{b,Rd}$; results CSV

4 · Worked example (numbers produced by the app engine)

W 110, D 90, t 2, lip 20, c 300, p 60, 2 × 10 mm web holes, 1 × 8 mm hole per flange, cap/base 130 × 100, t_{cap} 30, t_{base} 10, f_y 350, γ_{M0} 1.0, LC 0.01 mm/div, ΔN 1 kN. Lip and hole sizes are placeholders.

Quantity	Formula #	Alternative 1	Alternative 2
b	6	380 mm	380 mm
Flat widths b_w / b_f / b_l	7	108 / 88 / 19 mm	same
A_g / A_{net}	8, 11	644.0 / 572.0 mm²	same
λ_p web / flange / lip (at f_y)	12	1.160 / 0.945 / 0.623	same
ρ web / flange / lip	14, 15	0.698 / 0.812 / 1.000	same
A_{eff} (at f_y), Q	17	512.56 mm², 0.796	same
L_F / L_T	18	380 / 150 mm	300 / 300 mm
N_{cr}	19, 20	10,529 kN	4,488 kN
$\bar{\lambda}$ → χ	21, 22	0.131 → 1.000	0.200 → 1.000
$N_c = \chi A_{eff} f_y$	23	179.40 kN	179.40 kN
e	24 / 25	-1.265 mm	0
Failure load $N_u = R_k$	24 / 25, 31	**170.72 kN**	**179.40 kN**
Shortening / dial at failure	28	0.4656 mm / 46.6 div	0.5000 mm / 50.0 div
Proof load	29	67.36 kN	67.36 kN
Slope1 / 2 / 3 (kN/div)	30	3.966 / 3.495 / 3.667	3.966 / 3.393 / 3.588
$A_{eff} = R_k/f_y$	32	487.78 mm²	512.56 mm²
$N_{b,Rd} = R_k/\gamma_{M0}$	33	170.72 kN	179.40 kN

5 · Limitations and points to confirm

- Failure is calculated (formulas 23–25), then the loading loop stops at N_u; there is no falling branch after the peak — EN 15512 only records the maximum load.
- No initial imperfections or residual stresses are modelled, so the curve is straighter near failure than a physical test. Calibration against one real test is recommended.
- Flange-with-lip is treated as an internal plate (distortional buckling is excluded by the test purpose). Corner radii are ignored (square-corner centreline).
- Holes are taken in one cross-section (conservative). The hole-overlap rule (16) is a modelling choice.
- Alternative 2 is treated as concentric (e = 0). The 0.5 % holding load is described in the procedure but the app currently starts the readings at 0 kN.

- IS 2062 dropdown is reference only; the typed f_y drives every strength result. E, G, ν are fixed steel constants.
- EN 1993 clause numbers are quoted from knowledge — verify against your copies.